

WGFP_CombiningData_RShinyApp How-To

Location:

U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detctions\CodingDetctions\
WGFP_CombiningData_RShinyApp

Last updated: 7/8/2025

NOTE: If .txt files are not in the proper format, (as txt files downloaded from CU, CD, CS once were), include spaces in the appropriate places to combine data in the app. Use Edit_ Replace to edit.

Examples below:

- 00G to 00 G
- SCDANT to SCD ANT
- CD1A1 to CD1 A1

Uses

- Converting Stationary .txt files from ORSR readers to usable format and downloading it as .csv
- Combining Stationary .txt files from ORSR readers with Master Stationary .rds file
- Combining Biomark .xlsx files to master Biomark .rds file
- Combining other .csv files that have already been made/converted from .txt or .xlsx files

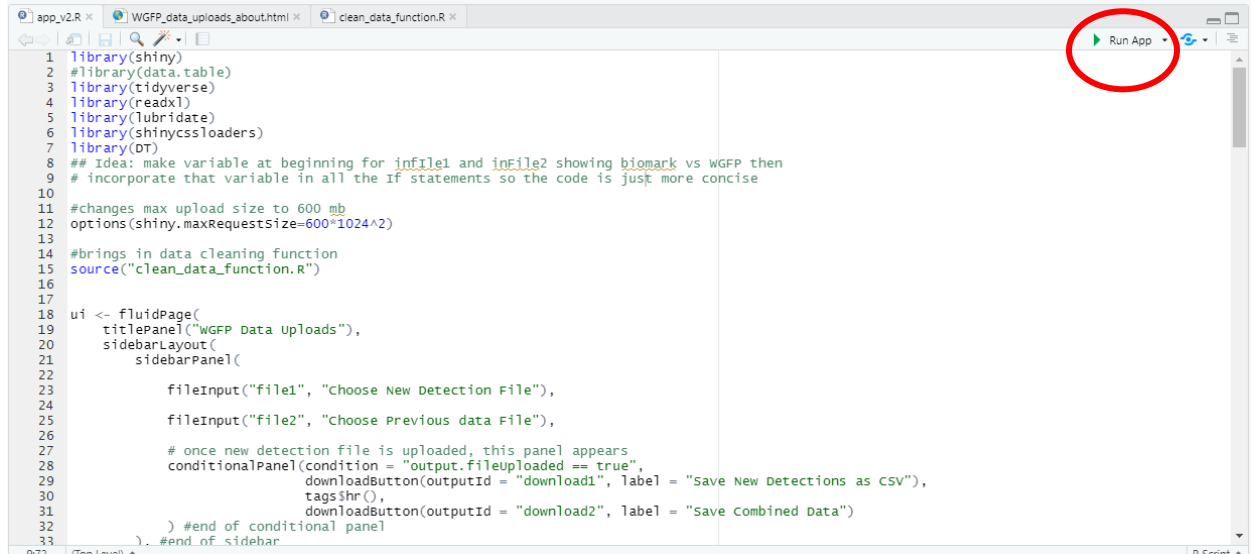
How to Open

OPENING FROM U: DRIVE

U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detctions\CodingDetctions\
WGFP_CombiningData_RShinyApp

1. Navigate to App location above and open the WGFP_updatedata_app "R Project" file. Rstudio will open
2. If the "app_v2.R" file isn't already open in the top left panel: in the bottom right panel, navigate to the "files" tab and open the app_v2.R file.

- Click “Run App” in the top right of the screen. You may be required to download some packages if this is the first time running the app.



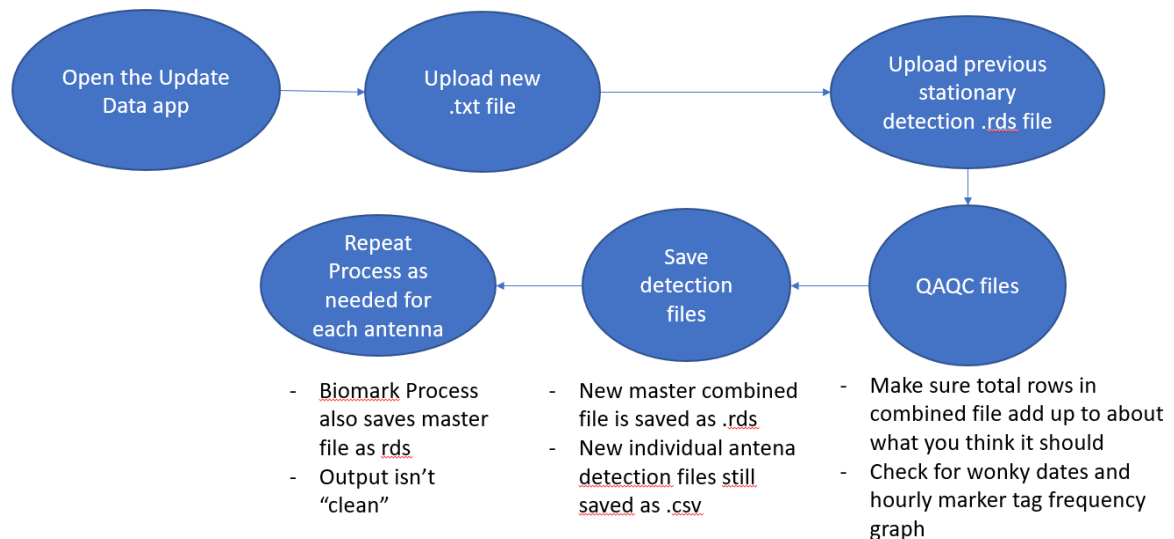
```

1 library(shiny)
2 #library(data.table)
3 library(tidyverse)
4 library(readxl)
5 library(lubridate)
6 library(shinycssloaders)
7 library(dt)
8 ## Idea: make variable at beginning for inFile1 and inFile2 showing biomark vs WGF then
9 # incorporate that variable in all the If statements so the code is just more concise
10
11 #changes max upload size to 600 mb
12 options(shiny.maxRequestSize=600*1024*2)
13
14 #brings in data cleaning function
15 source("clean_data_function.R")
16
17
18 ui <- fluidPage(
19   titlePanel("WGFP Data Uploads"),
20   sidebarLayout(
21     sidebarPanel(
22       fileInput("file1", "Choose New Detection File"),
23       fileInput("file2", "Choose Previous data File"),
24
25       # once new detection file is uploaded, this panel appears
26       conditionalPanel(condition = "output.fileUploaded == true",
27         downloadButton(outputId = "download1", label = "Save New Detections as CSV"),
28         tags$hr(),
29         downloadButton(outputId = "download2", label = "Save Combined Data")
30       ) #end of conditional panel
31     ),
32     . #end of sidebar
33   )

```

How to Use

Overview: Upload new detections, upload previous/master file of detections, QAQC both those files, then download new detections and new combined file as CSV's in appropriate directories. Then repeat as needed for each new detection file.



- Click “Browse” under “Choose New Detection File and choose the most recent detection file from the U: Drive, located under
U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detections\(**STATION ARY SITE**)

WGFP Data Uploads

Choose New Detection File

Browse... No file selected

Choose Previous data File

Browse... No file selected

How to Use

New Detections

Previous Detections

Combined Data

How to Use

Upload the most recent data files from the Biomark or Stationary Antennas in .txt (Stationary), .csv (Stationary file that has already been combined), or .xlsx (Stationary file that has already been combined). Upload the previous combined WGFP_Raw or Biomark_Raw.csv using "Choose Previous Data File". Then, click "Combined Data" tab. Save the New Detection File as its own CSV with "Save New Detections as CSV", and click "Save Combined Data" and save. Then, choose a new file to upload and re-upload the previously combined WGFP_Raw or Biomark_Raw file. Reupload the new file.

Notes:

Older .txt files from the beginning of the project won't currently work because they have different fields than the current fields being used. .txt files are automatically cleaned to a viewable, usable form which often takes some time. A 10 mb file should take about 27 minutes. If the new detections file is too large it may not display on the table.

For this example, we'll use a .txt file as obtained directly from the Oregon ORSR readers around Windy Gap. Here is an example file of what the .txt file should look like to be uploaded in the app. See QAQC section for incorrect example file

```

CRCC_CD_A8_20250331 - Notepad

File Edit View

up
Reader: A8 Site: CD2
----- Upload #12, 2398 records -----
DTY      ARR      TRF      DUR      TTY      TAG      SCD ANT      NCD      EFA
E 2025-03-11 11:53:30.322 G End of upload #11
S 2025-03-11 12:01:24.600 G 00:00:02.600 A 900_230000087405 CD2 A1 27 0.6
S 2025-03-11 12:01:33.600 G 00:00:01.000 A 900_230000087405 CD2 A1 11 0.7
S 2025-03-11 12:01:46.500 G 00:00:00.500 A 900_226001581072 CD2 A1 6 0.5
S 2025-03-11 12:01:50.300 G 00:00:00.000 A 900_226001581072 CD2 A1 1 0.8
S 2025-03-11 12:01:56.600 G 00:00:00.700 A 900_226001581072 CD2 A1 8 0.8
S 2025-03-11 12:02:19.100 G 00:00:00.000 A 900_230000087405 CD2 A1 1 0.9
S 2025-03-11 12:02:25.200 G 00:00:01.800 A 900_230000087405 CD2 A1 19 1.0
S 2025-03-11 12:02:28.800 G 00:00:00.000 A 900_230000087405 CD2 A1 1 1.1
S 2025-03-11 12:02:37.500 G 00:00:00.800 A 900_230000087405 CD2 A1 9 0.7
S 2025-03-11 12:02:48.600 G 00:00:01.200 A 900_226001581072 CD2 A1 13 0.9
S 2025-03-11 12:02:51.000 G 00:00:00.000 A 900_226001581072 CD2 A1 1 0.7

```

- Once the upload is complete, the data will be cleaned into a viewable format using the function `clean_txt.r`. Depending on how big the new .txt detection file is, this can take quite some time. A .5 mb file usually takes 20-30 seconds, 10 mb file can take about 3 minutes, and a 30 mb file took 27 minutes.

If it's a stationary file, the data then is cleaned using the `cleanStationary.r` function. This function removes underscores and trims whitespace from the TAG column, standardizes the date format in the DTY column to "yyyy-mm-dd", cleans the ARR timestamps by handling AM/PM formats and removing 10ths of seconds, filters rows to ensure valid date entries (occurred after 2020-08-06) and specific Code values ("I" and "S"), and removes duplicate rows.

After the data is cleaned, it will show up in the "New detections" tab.

WGFP Data Uploads

Choose New Detection File

Browse...
CRCC_CD_A8_202401

Upload complete

Choose Previous data File

Browse...
WGFP_Raw_20240514

Upload complete

Save New Detections as CSV

Save Combined Data

How to Use

New Detections

Previous Detections

Combined Data

QAQC

Show 10 entries
Search:

	Code	DTY	ARR	TRF	DUR	TTY	TAG
1	S	2023-11-28	13:21:24	G	00:00:00.100	A	900230000087408
2	S	2023-11-28	13:21:27	G	00:00:03.600	A	900230000087408
3	S	2023-11-28	13:21:32	G	00:00:00.000	A	900230000087408
4	S	2023-11-28	13:21:44	G	00:00:00.400	A	900226001546996

- After the file is uploaded, do some quick QAQC by looking over the file in the New Detections tab, including checking that marker tags were detected normally, the “ARR” column was read in correctly, “SCD” column is correct, there are about as many entries in the table as you’d expect to see, and the Dates are from the last date the sites were downloaded to the most recent date. For more info, see the QAQC section.
- Upload previous detections using the “Choose Previous data File” input.
U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detections\All_Stationary. File starts WGFP_Stationary_Raw. For Biomark Antennas, this file is located at U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detections\Biomark. Once this is uploaded, it will show up in the Previous Detections Tab

WGFP Data Uploads

Choose New Detection File

Browse...
CR_RB_A1_20220110

Upload complete

Choose Previous data File

Browse...
No file selected

Save New Detections as CSV

Save Combined Data

How to Use

New Detections

Previous Detections

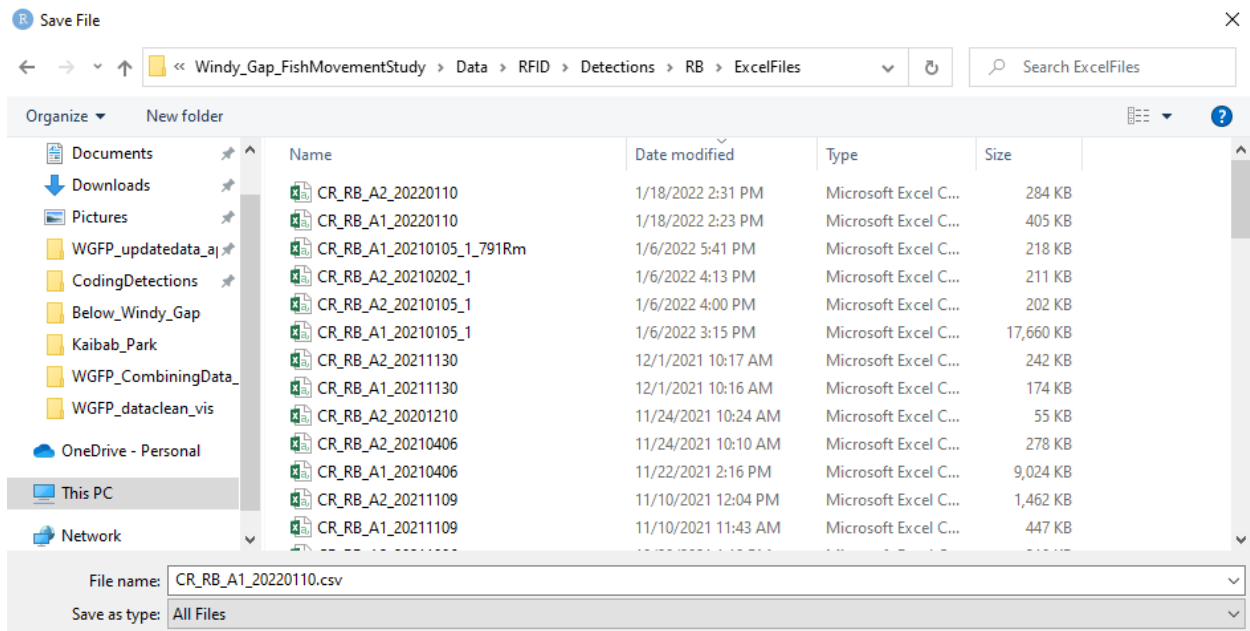
Combined Data

QAQC

Show 10 entries
Search:

	Code	DTY	ARR	TRF	DUR	TTY	TAG
2	S	2021-11-30	11:17:21.900	G	00:00:03.400	A	900_23000010275
3	S	2021-11-30	11:17:29.600	G	00:00:00.000	A	900_23000010275
4	S	2021-11-30	11:17:38.800	G	00:00:02.600	A	900_23000010275
5	S	2021-11-30	11:17:43.200	G	00:00:01.800	A	900_23000010275
6	S	2021-11-30	11:17:49.400	G	00:00:02.600	A	900_23000010275

- Do some QAQC here too, particularly making sure the “ARR” column is read in correctly. When read in correctly, it will contain hours, minutes, and seconds. Biomark data will include tenths of seconds, but as of 2024, Stationary data does not. Example: “13:16:05”. See QAQC section for more details.
- Click “Save New Detections as CSV”. Navigate to U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detections\(**STATIONARY SITE**)\ExcelFiles, add “.csv” to the end of the filename, and save.



- Go to the "Combined Data" Tab, which combines what is uploaded as "New Detections" with "Previous Detections". Check that ARR column looks Ok, the number of entries look ok, and DTY ends with the most recent date. In this case, there were 5,114 entries in the new_detections file, and 2681957 entries in the "Previous Detections" file. So you would expect to see 2687071 entries in the Combined Data tab, shown in the row number at the top left of the screen.

WGFP Data Uploads

Choose New Detection File

Browse... CRCC_CU_A11_20250331.TXT

Upload complete

Choose Previous data File

Browse... WGFP_Raw_20241204_c08.nds

Upload complete

Save New Detections as CSV

Save Combined Data

How to Use New Detections Previous Detections Combined Data QAQC

Number of Rows: 2,296

Show 10 entries

Code	DTY	ARR	TRF	DUR	TTY	TAG	SCD	ANT	NCD	EFA
1	S	2025-03-11 12:34:21	G	00:00:05.300	A	900230000087405	CU1	A1	54	0.8
2	S	2025-03-11 12:34:28	G	00:00:00.000	A	900230000087405	CU1	A1	1	0.8
3	S	2025-03-11 12:34:35	G	00:00:00.100	A	900230000087405	CU1	A1	2	0.8
4	S	2025-03-11 12:34:56	G	00:00:00.000	A	900226001581072	CU1	A1	1	0.8
5	S	2025-03-11 12:35:01	G	00:00:00.100	A	900226001581072	CU1	A1	2	0.8
6	S	2025-03-11 12:35:27	G	00:00:02.100	A	900230000087405	CU1	A1	22	0.8

3 0.8

14 0.8

4 0.8

10 0.9

5 ... 230 Next

How to Use New Detections Previous Detections Combined Data QAQC

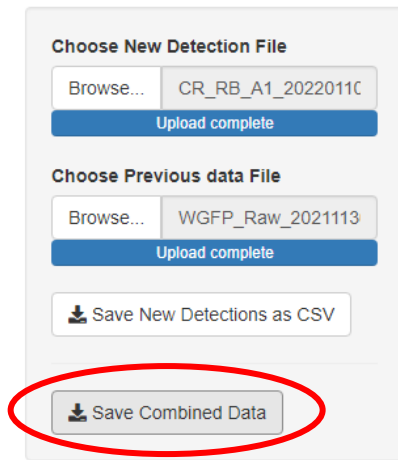
Number of Rows: 2,871,430

Show 10 entries

Code	DTY	ARR	TRF	DUR	TTY	TAG
1	S	2021-03-02 13:01:11	G	00:00.9	W	0000
2	S	2021-03-02 13:16:06	G	00:00.9	W	0000
3	S	2021-03-02 13:22:41	G	00:00.0	R	0000
4	S	2021-03-02 13:23:55	G	00:00.0	R	0000
5	S	2021-03-02 13:24:48	G	00:00.0	R	0000

- a. Note: sometimes the row count located below the table won't update. Because of this, as of October 2024, a correct row counter was added at the top of the table to accurately reflect the number of rows, even if the row counter at the bottom of the table doesn't update.
8. Check the QAQC tab for relevant QAQC, including checking for Problem times, daily Marker tag distributions, and overall marker tag distributions. See QAQC section.
9. Click Save Combined Data and Save as "WGFP_Stationalary_Raw_YYYYMMDD_RB1.rds" in U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detctions\All_Stationalary. This will become your next "Previous Detctions" file to add the next "New Detctions" file onto.

WGFP Data Uploads



Choose New Detection File

Browse... CR_RB_A1_20220110

Upload complete

Choose Previous data File

Browse... WGFP_Raw_2021113

Upload complete

Save New Detctions as CSV

Save Combined Data

10. Repeat the process over with the next .TXT file, this time using "WGFP_Stationalary_Raw_YYYYMMDD_RB1.rds" as the "Previous Detctions" file. If using the app online, refresh the app to clear any caching.
11. Repeat the process over again, saving each new Combined file suffixed with _RB2, _HP3, _HP4, _CF5, _CF6 until you're finished with all the new files. Each of these files should be a bit larger than the previous file.
12. Rename "WGFP_Stationalary_Raw_YYYYMMDD_CF6.rds" (assuming this file now contains all new detctions from RB1 through CF6) to WGFP_Stationalary_Raw_YYYYMMDD.rds. Delete or archive previous iterations ending in _SiteCode. Move previous month's Master Combined file to Archive.

QAQC

QAQC Tab

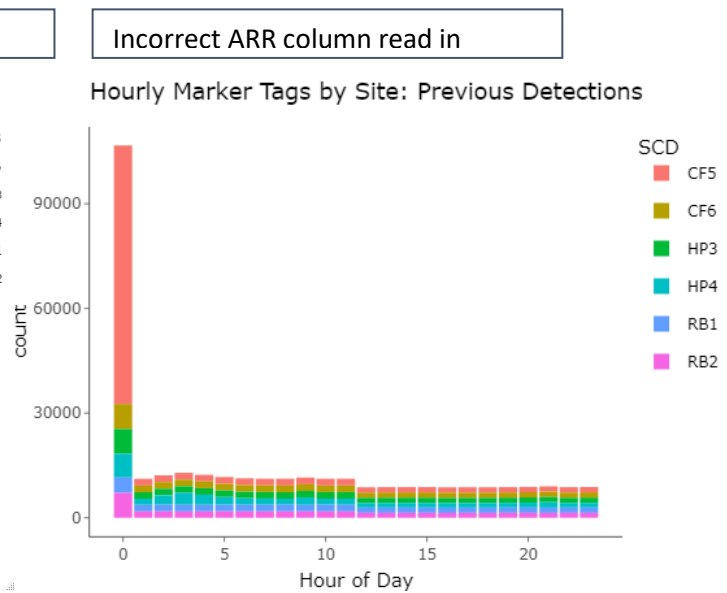
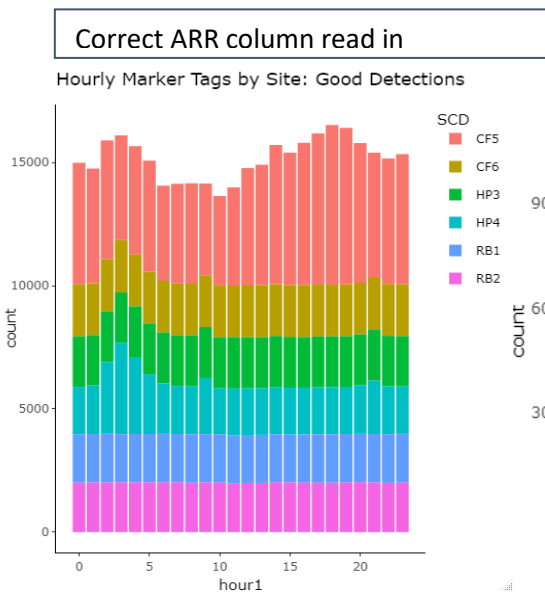
Problem Times and Hourly Marker Tag

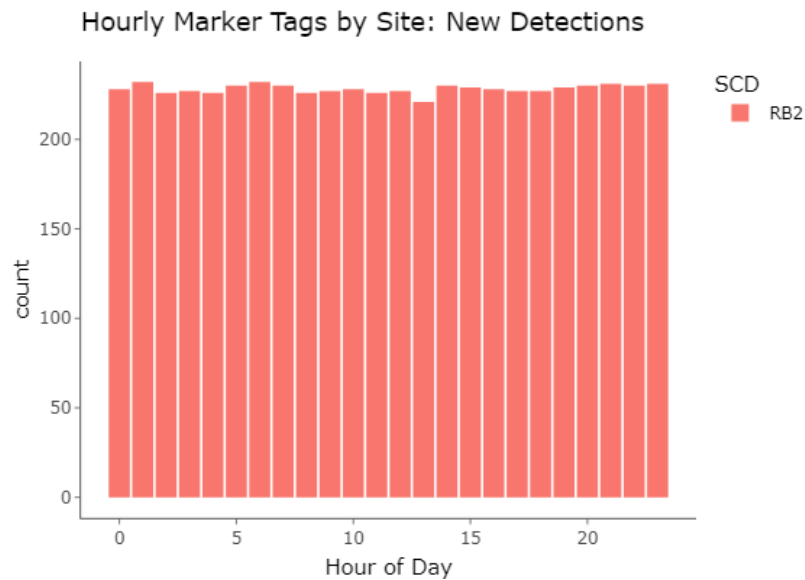
The datatable in the QAQC Tab keeps track of "problem times", where ARR was read in incorrectly. If all times are read in correctly, this table will be blank.

- Check that the ARR column of new detections and previous detections shows up in the format HH:MM:SS (Biomark will have 10ths of a second, Stationary files will not). If not, the problem rows will show up in the table. It is a good thing for this table to blank
 - This is when the ARR column gets corrupted and can happen sometimes seemingly randomly, mainly in the previous detections file. This screened for in a couple different ways. The first way is that you can see it show up in the datatable.

Correct	Incorrect
ARR	ARR
13:21:24	01:10.8

- ARR is cleaned into a Datetime format YYYYMMDD HH:MM:SS, but when ARR is brought in incorrectly, it will get translated with hour 0. In the incorrect example above, the time would get translated as 00:01:10, or 12:01 AM and 10 seconds. This error will then show up in the frequency hourly bar plots in the “Hourly Marker Tag” tab as shown below with an excessive number of detections occurring between 12 and 1 AM. Marker tags only are plotted to make the plot rendering process quicker.





1 If the ARR field (or in Biomark case, Scan Time) isn't corrupted or has bad data, this is what the frequency plot of marker tags should look like: pretty equal distribution of detections across all hours of the day

- The ARR column could be brought in as blank. Check Previous Detections and New Detections files to make sure this is not the case.
- If you find problems with the ARR column, the easiest way to “fix” this is to return to a previous copy of the Master Detection file without problem times and start over from there.

Overall Marker Tags

Marker tags from New Detections are parsed and graphed as a scatter plot. This is helpful to see instances where antennas or marker tags were not functioning properly. This is a small mirror of the QAQC -> Marker Tags tab in the WGFP Data Visualization app. If there are significant downtimes (typically more than 1 day of no marker tag detections), then this should be recorded in the WGFP Metadata, found at

U:\Projects\Colorado_River\Windy_Gap_FishMovementStudy\Data\RFID\Detections

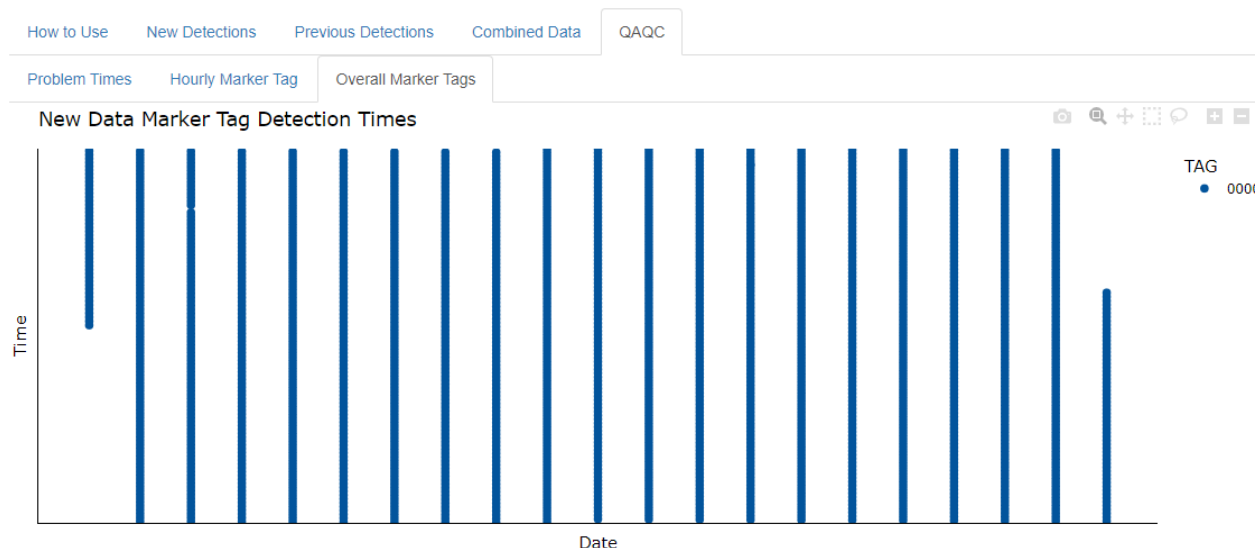


Figure 2 This is a normal distribution of marker tag detections throughout the period of New Detections

Other QAQC Checks

When combining the data, there are a few things to check for to make sure a good dataset is being made.

- .TXT files being uploaded are > 1 kb
 - This means that the data that was downloaded from the readers has actual detections.



- Rows at the end of the file get automatically removed, but if there are more rows at the beginning of the file such as the screenshot below, the file won't parse correctly and 0 rows will show in the "New detections" tab.

```

CR_CF_A5_20241107 - Notepad
File Edit View

up
rup

HSG*> up
Reader: rup Site: CF5
----- Upload #7, 4591 records -----
DTY      ARR      TRF      DUR      TTY      TAG      SCD ANT      NCD      EFA
E  2024-10-02 15:47:35.944 G End of upload #6
S  2024-10-02 15:53:55.900 G 00:00:01.400 A 900_230000087405 CF5 A1 15 1.2
S  2024-10-02 15:54:26.800 G 00:00:01.300 A 900_230000087405 CF5 A1 14 1.1
S  2024-10-02 15:54:30.000 G 00:00:00.400 A 900_230000087405 CF5 A1 5 1.2
S  2024-10-02 15:54:35.400 G 00:00:00.000 A 900_230000087405 CF5 A1 1 1.2
S  2024-10-02 15:54:37.000 G 00:00:01.000 A 900_230000087405 CF5 A1 14 1.2

```

- Check that when adding each new site to the master file, the new combined file is bigger in size than the previous combined file. If it's smaller, the most likely culprit is that the ARR column got corrupted and you need to re-combine that previous detection.

- With the Biomark antennas, sometimes one file downloaded can contain duplicated info from the last Biomark file that was downloaded. This is checked for in excel by seeing if there are multiple ReaderID entries in a new biomark file.
 - Check to make sure that this in fact is duplicated data by opening the other file where you suspect contains the source of the duplicated data and cross-check to see if this is the case.
 - This isn't a big deal because duplicated data gets cleaned later before analysis, but it is helpful to be aware of rows that will be deleted in the Master File.
- Check that file type is the type you'd expect, either .csv or .rds. Sometimes you need to add ".csv" to the names of the new file, and add ".rds" to the combined files that you save.
 - If you forget, there will be a blank file type in file explorer. you can go back and add ".csv" or ".rds" in file explorer and that will change it to the correct file type.